Video understanding. 视频理解.

2D + Time:

由一组图像组成：4D张量

$T \times 3 \times H \times W$

(时间).

Example task: Video Classification.



model → Swimming / Running / Eating ...

① 图像：理解 "objects"：Dog. Cat. Truck ...

视频：理解 "actions".

② Videos are big!

30帧. 60帧 ...

SD (640×480)：~1.5GB per minute ⇒ 无法直接输入GPU/模型.

HD (1920×1080)：~10GB per minute.

Simple Solution：压缩图像.

减少长.宽 (空间分辨率)

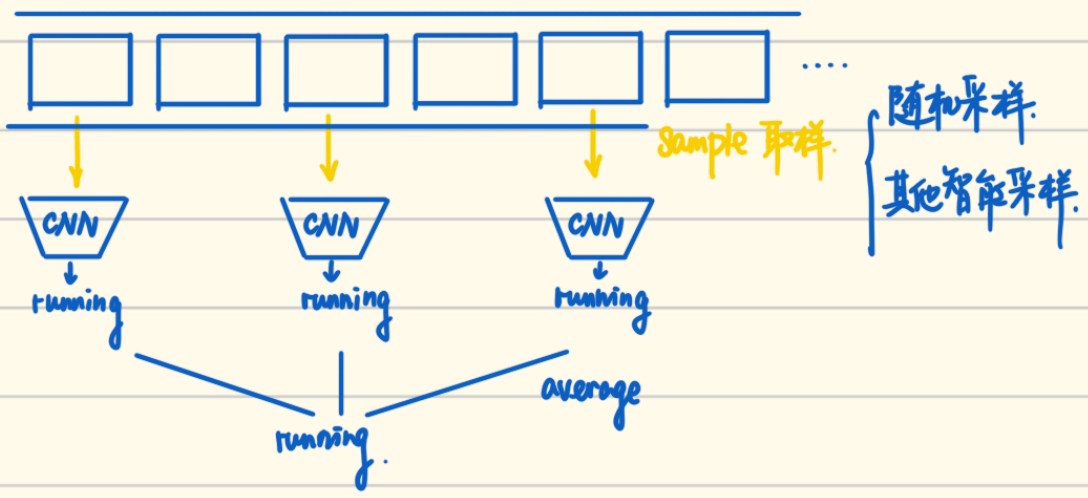
减少帧 (时间).

Train on Clips.

采样一段视频的片段，在不同模型上运行，以平均值作为结果

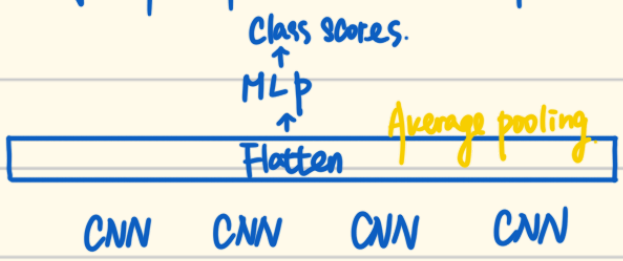
Single-Frame CNN:

简单地将视频当作多个图像，放入CNN预测并取平均.



Late Fusion (with FC Layers).

对采样图像使用2D CNN，提取特征向量，扁平连接，最后用MLP分类.



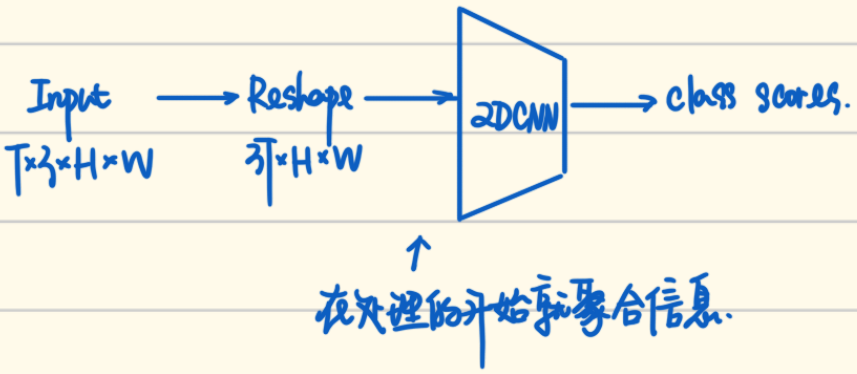
另一种想法：全局池化.

(减少全连接的维度).

但也会损失信息.

缺点：直觉来说，在最后环节聚合可能会丢失"底层信息".

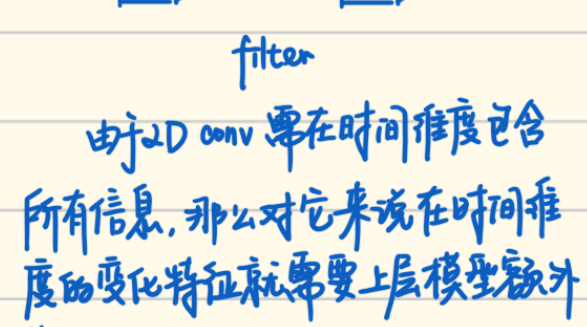
Early Fusion



缺点：用单个神经网络在单层捕获所有内容信息被折叠.

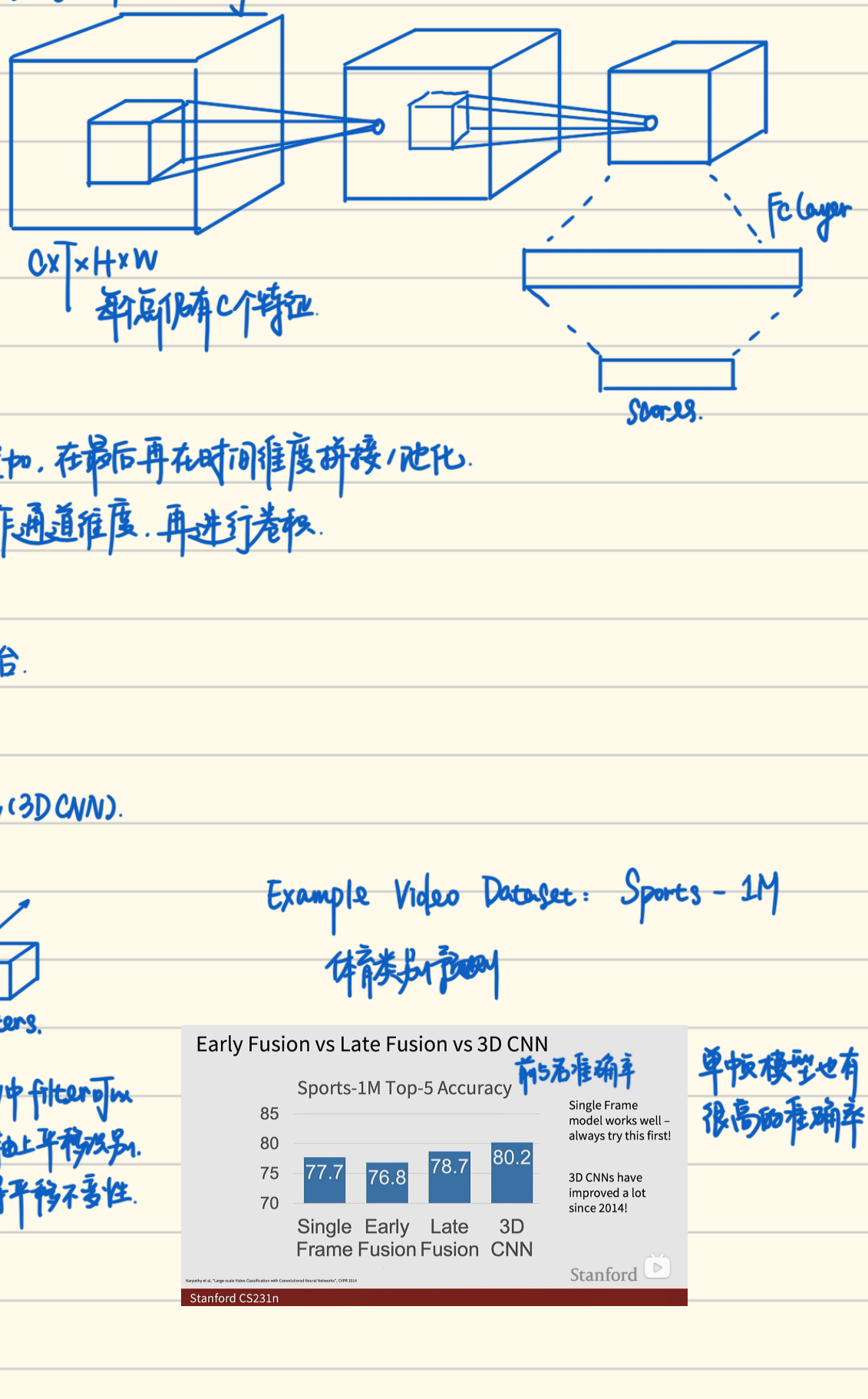
↑ 在处理的开始就聚合信息.

3D CNN：中间的聚合.



在整个CNN中，信息缓慢聚合.

3D Convolution Layer.



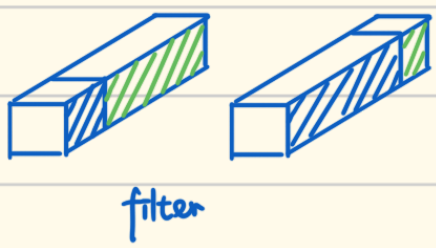
对Late Fusion，聚焦在特征维度持续叠加，在最后再在时间维度拼接/池化.

Early Fusion：先在时间维度折叠，视作通道维度，再进行卷积.

⇓

3D CNN：在空间与时间上都缓慢融合.

2D Conv (Early Fusion) VS 3D Conv (3D CNN).



filter

由于2D conv 聚在时间维度已合所有信息，那么对它来说在时间维度的变化特征就需要上层模型额外学习，但这也会导致上层模型不能得到一个"普遍"的时间维特征。

失去平移不变性.



filters.

3D CNN中filter可以在时间轴上平移以习得平移不变性.

Example Video Dataset: Sports-1M

体育类别数据集

单帧模型也有很高的准确率

C3D: The VGG of 3D CNNs. 类似于2D情形下的VGG.

C3D: The VGG of 3D CNNs

3D CNN that uses all 3x3x3 conv and 2x2x2 pooling (except Pool1 which is 1x2x2)

Released model pretrained on Sports-1M: Many people used this as a video feature extractor

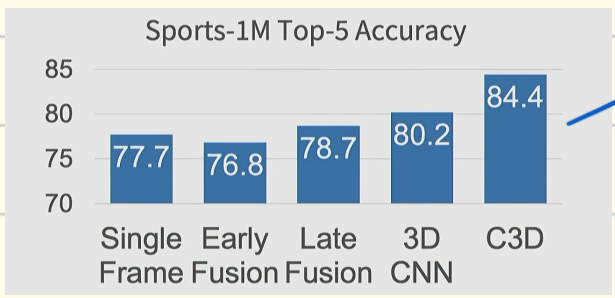
Problem: 3x3x3 conv is very expensive! 缺点：非常昂贵.
AlexNet: 0.7 GFLOP
VGG-16: 13.6 GFLOP
C3D: 39.5 GFLOP (2.9x VGG!)

Layer	Size	MFLOPs
Input	3 x 16 x 112 x 112	
Conv1 (3x3x3)	64 x 16 x 112 x 112	1.04
Pool1 (1x2x2)	64 x 16 x 56 x 56	
Conv2 (3x3x3)	128 x 16 x 56 x 56	11.10
Pool2 (2x2x2)	128 x 8 x 28 x 28	
Conv3a (3x3x3)	256 x 8 x 28 x 28	5.55
Conv3b (3x3x3)	256 x 8 x 28 x 28	11.10
Pool3 (2x2x2)	256 x 4 x 14 x 14	
Conv4a (3x3x3)	512 x 4 x 14 x 14	2.77
Conv4b (3x3x3)	512 x 4 x 14 x 14	5.55
Pool4 (2x2x2)	512 x 2 x 7 x 7	
Conv5a (3x3x3)	512 x 2 x 7 x 7	0.69
Conv5b (3x3x3)	512 x 2 x 7 x 7	0.69
Pool5	512 x 1 x 3 x 3	
FC6	4096	0.51
FC7	4096	0.45
FC8	0	0.05

Tran et al, "Learning Spatiotemporal Features with 3D Convolutional Networks", ICCV 2015

Stanford CS231n

GFLOPs：一次forward所需要的浮点计算数，用于衡量效率.



提升3到4%准确率

运动与外观：其特征不相似，何不尝试用不同网络？

Measuring Motion: Optical Flow. 光流

反映像素在相邻帧的运动.

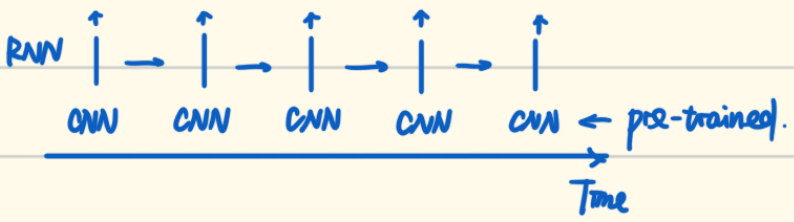
⇓

测量像素在相邻帧中的运动.

Separating Motion and Appearance: Two-Stream Network. 双流网络.



Modeling long-term temporal structure.

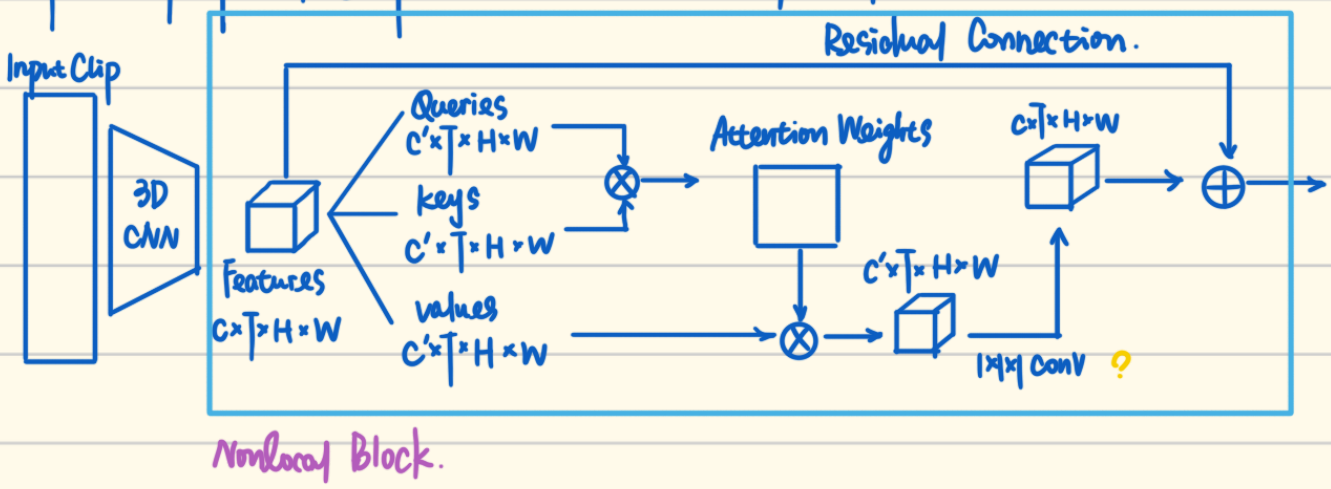


RNN缺陷：不并行.

↓

Self-attention

Spatio-Temporal Self-Attention (Nonlocal Block).



Nonlocal Block.

Inflating 2D Network to 3D (I3D)

使用2D架构，作用于video（使用原先模型的预训练数据）.

Temporal Action Localization